

Lab 1: Requirements Engineering & UML Use-Case Modelling

Use-Case Flow Specification

Use Case: UC-03 — Generate Rubric Score Breakdown

Primary Actor: Faculty Evaluator

Preconditions:

1. The Faculty Evaluator has configured a valid rubric template and unit test suite for the assignment.
2. A student project submission has been uploaded and successfully added to the evaluation queue.
3. The required evaluation tests are available for execution.

Postconditions:

Success:

- The configured test suite has been executed.
- Test results have been recorded.
- The rubric has been applied to the test results.
- An itemized rubric score breakdown has been generated and stored.
- The submission is automatically assigned for peer review.

Failure:

- If the submission cannot be tested successfully, no invalid final score is generated.
- The failure information is recorded and the submission is marked as invalid.

Main Success Scenario

1. Faculty Evaluator selects a queued student submission for evaluation.
2. System retrieves and validates the submitted project.
3. System executes the configured syntax checks and unit test suite.
4. System records the pass/fail result of each individual test case.
5. System maps the test results to the configured rubric criteria.
6. System calculates the score for each rubric criterion.
7. System calculates the total rubric score and generates the itemized score breakdown.
8. System stores the score breakdown and test results against the submission.
9. System automatically assigns the successfully tested submission to two eligible, anonymized peer reviewers.
10. System makes the evaluation result available to the Faculty Evaluator and Student.
11. The use case ends successfully.

Alternate Flow — Test Suite Execution Failure

A1. Test Execution Failure

1. During test execution, the system detects a build, compilation, or test execution failure.
2. The system records the specific error and failed test information.
3. The system marks the submission as **“Invalid – Build/Test Failed.”**
4. The system does not generate an invalid final rubric score.
5. The Student is notified of the failure and provided with the relevant error information.
6. The Student may correct the project and resubmit it before the submission deadline.
7. If the submission is resubmitted, the evaluation process starts again.